

Understanding Financial Well-Being Across Age Groups in the U.S

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Contribution: Ebenezer Ampofo: Visualization, Summarization

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Both: Research, Writing, Insights

Project Goals:

The primary goal of this project is to understand how financial well-being differs across age groups and to identify the economic and demographic factors that drive these differences. Financial well-being is closely tied to an individual's ability to manage expenses, handle financial shocks, and feel secure about long-term stability, making it an important area of analysis. For college-aged adults, financial constraints often represent themselves in lower income, limited savings, and inconsistent employment, while older adults may experience different pressures, such as long-term debt or greater household responsibilities.

Using data from the Consumer Financial Protection Bureau's National Financial Well-Being Survey, we examine how variables such as income, employment, education, savings behavior, budgeting habits, and credit use relate to financial well-being scores across different age groups. This dataset, in particular, is well-suited to our research because it provides a validated financial well-being score alongside a wide range of economic, behavioral, and demographic attributes, allowing us to explore both structural and behavioral drivers of financial well-being. Through statistical summarization, visualization, association analysis, predictive modeling, and clustering, our goal is to uncover meaningful patterns that explain why certain age groups experience greater financial strain and which factors most strongly contribute to differences in financial well-being.

Dataset Description & Preprocessing

The dataset chosen comes from the 2016 National Financial Well-Being Survey, which includes a validated financial well-being score (FWBscore) along with variables related to income, employment, savings habits, budgeting behavior, credit use, household structure, and demographic characteristics. These attributes allow for meaningful comparisons of financial behaviors and outcomes.

To prepare the data for analysis, we transformed several survey-coded variables into usable formats. Income categories were converted into midpoint estimates (income_mid). Measured behavioral variables on Likert scales, such as budgeting, credit repayment, and saving frequency, were recorded into binary indicators using a threshold of ≥ 4 . The emergency savings variable was similarly recorded, with values ≥ 3 treated as indicating the presence of meaningful savings.

We also created an age-group variable distinguishing young adults (18–24) from older adults (25+) and generated indicators for student status, part-time employment, and the number of household earners. Rows missing the financial well-being score were removed, since this is the outcome variable. Other missing values were handled by converting refusal codes to standard missing markers and applying median imputation for modeling purposes.

Table of Key Variables:

Variable Name	Original Codebook Name	Meaning	Type
FWBscore	FWB1	Financial well-being score (0-100)	Continuous
young_adult	AGE	1 = age (18-24), 0 = (25+)	Binary
income_mid	PPINCIMP	Midpoint of household income range	Continuous
budget_adherent	MANAGE1_2	Reports adhering to a budget (Likert ≥ 4)	Binary
credit_paid_full_monthly	MANAGE1_3	Pays credit card in full monthly (Likert ≥ 4)	Binary
save_habit_good	SAVEHABIT	Saves regularly (Likert ≥ 4)	Binary
emergency_savings_present	SAVINGSRANGES	Reports meaningful emergency savings (≥ 3)	Binary
is_student	STUDENT	Respondent is a student (1 = yes)	Binary
is_part_time	EMPLOYMENT	Part-time employment	Binary
EARNERS_clean	EARNERS	Number of earners in household	Numeric

Data Wrangling & Assembly

Additional indicators were created for student status, part-time employment, and the number of household earners. Rows missing the FWBscore were removed, as this is the outcome variable for several analyses, and refusal codes were converted to standard missing values. For modeling

tasks, the remaining missing predictors were filled using median imputation to maintain consistency.

After these steps, we assembled the final analysis dataset (model_df), which includes the financial well-being score and the behavioral, economic, and demographic variables used throughout the project.

Statistical Summary

Older adults report higher financial well-being on average (≈ 56.4 vs. 50.8) and higher estimated income levels ($\approx 88k$ vs. $67.5k$).

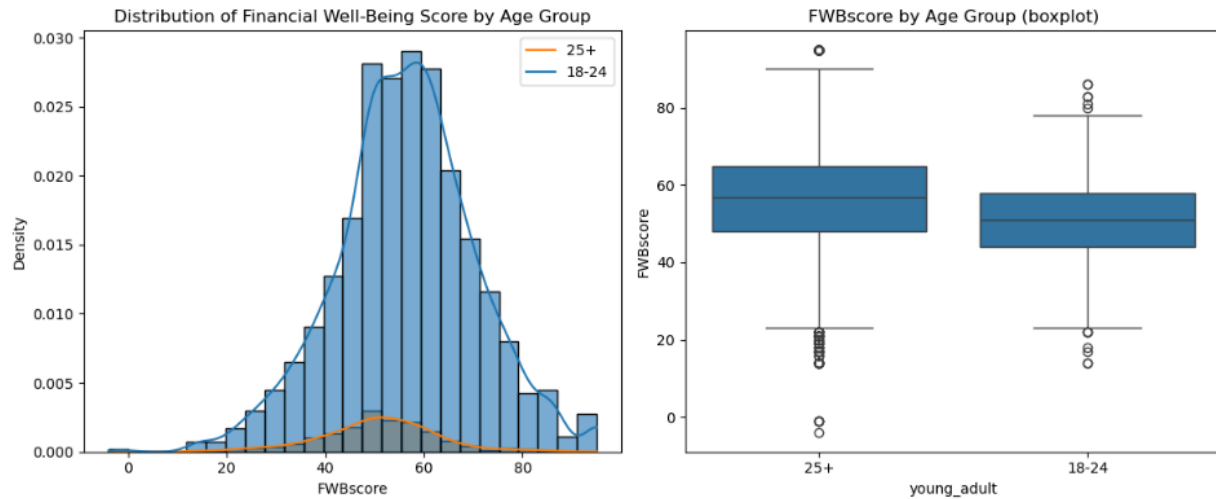
They also show stronger financial habits:

higher rates of emergency savings, budgeting, saving regularly, and paying credit balances in full. Indicators of financial resilience similarly favor older adults.

These differences align with the **t-test** result ($t = -9.24$, $p \approx 7e-19$), confirming a statistically significant gap in financial well-being between the two groups. Overall, the summaries suggest that lower income and less consistent financial habits may help explain why young adults experience weaker financial well-being, motivating the deeper analyses in later sections.

Variable	25+ Mean	18–24 Mean
FWBscore	56.4	50.8
Income (midpoint)	88k	67.5k
Emergency savings (%)	87.6	79.0
Budget adherence (%)	68.0	58.2

Visualization



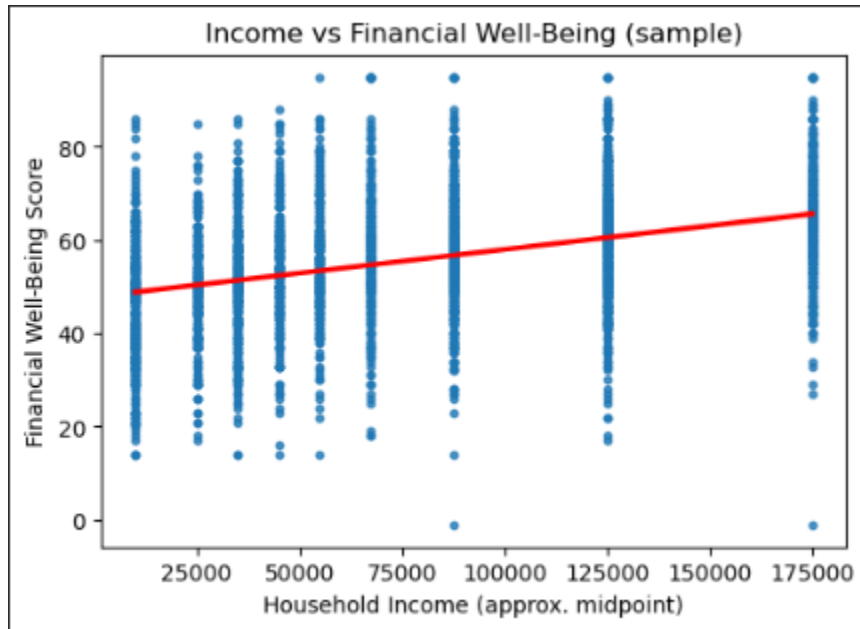
Histogram + KDE Distribution of FWBscore by age group

The histogram with KDE overlay shows that both groups follow a roughly normal distribution, but the young adult distribution is visibly shifted lower, indicating overall lower scores. The density curve for the 18–24 group is flatter and slightly wider, suggesting more variability in financial well-being within that age range.

Boxplot of FWBscore by age group

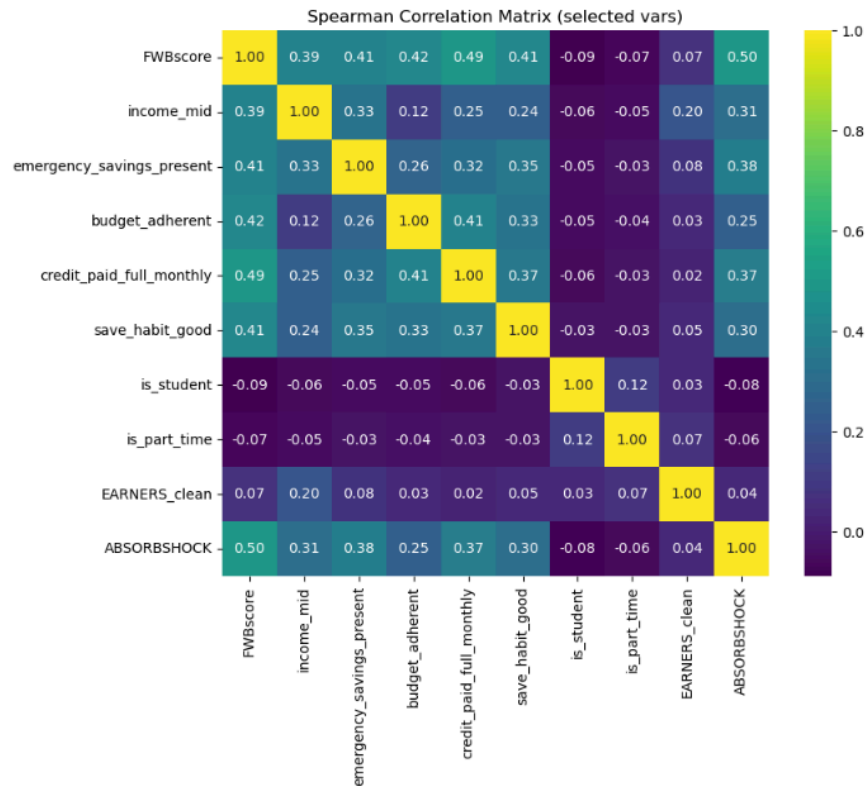
Older adults display a higher median FWBscore and a more compact interquartile range, consistent with more stable financial standing. In contrast, the young adult group exhibits a lower median and a greater presence of low-end outliers, reflecting more frequent financial strain. These visual trends align with the descriptive statistics and suggest meaningful age-based disparities in financial well-being before formal statistical testing.

Association Analysis



The income-FWB score scatterplot shows a clear positive trend: individuals with higher income tend to report higher financial well-being, though there is wide variation at each income level.

Using a Spearman correlation matrix, we examined relationships between financial well-being and selected variables. Budgeting, saving habits, emergency savings, and paying credit balances in full all show moderate positive associations with FWBscore, while student status and part-time work show small negative associations. Overall, both financial behaviors and economic capacity are closely related to financial well-being.



T-Test: Comparing Financial Well-being by Age Group

We conducted an independent samples t-test to compare FWBscore between young adults (18–24) and adults 25+.

The results show a significant difference between the two groups:

t-statistic: -9.239

p-value: 7.04e-19

Because the p-value is far below 0.05, we reject the null hypothesis. Young adults report significantly lower financial well-being than older adults on average. This supports our expectation that income constraints, limited savings, and student status contribute to lower well-being among younger respondents.

Predictive Modeling

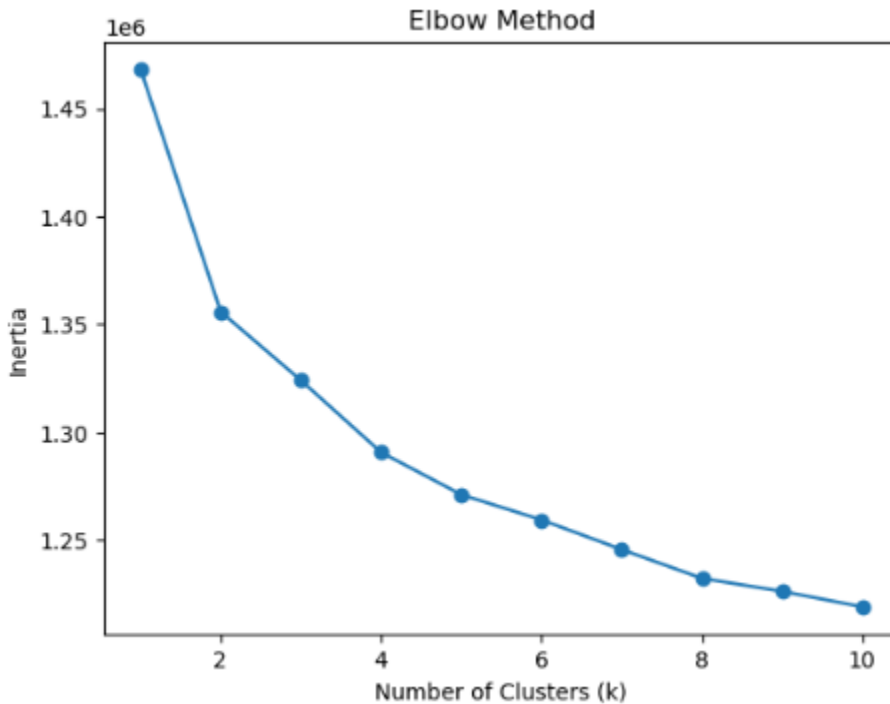
Linear Regression $R^2 \approx 0.394$

RMSE ≈ 10.919

The model explains about **39%** of the variation in financial well-being. Budgeting, saving

behavior, paying credit bills on time, income, and emergency savings emerge as the strongest predictors of a higher FWBscore.

Clustering



K-Means was used to explore whether individuals group into distinct financial well-being profiles. The elbow method indicates that $k = 3$ is a reasonable choice, balancing simplicity and model fit.

With three clusters, the data separates into:

- a higher-income, higher-FWB group with strong financial habits,
- a moderate group with mixed behaviors, and
- a lower-income, lower-FWB group with weaker habits.

These clusters align with earlier findings: young adults appear more concentrated in the lower-FWB cluster, reinforcing the patterns observed in the statistical summary and t-test.

Challenges

One major challenge we encountered involved missing values in the dataset, which likely resulted from survey respondents skipping certain questions. Although we initially removed missing values during preprocessing, we continued to encounter errors when running the predictive models, indicating that some missing values still remained. To address this, we applied an additional “dropna” step within each predictive modeling workflow, which resolved the issue and allowed the models to run successfully.

Insights

Across all analyses, financial well-being is shaped by both structural factors (income, employment, savings capacity) and behavioral patterns (budgeting, saving, and credit management). Young adults, especially students and part-time workers, consistently score lower on these dimensions, which helps explain their lower average financial well-being. While income plays an important role, strong financial habits are also closely associated with higher well-being across age groups, suggesting that behavior matters alongside economic constraints.

Future Directions

Future work could incorporate the survey’s weighting variable to produce population-representative estimates, which may refine group comparisons. Additional behavioral or psychological measures, such as financial confidence or risk tolerance, could also help explain the remaining variance in financial well-being. Expanding the analysis to include multiple years of the survey or testing alternative clustering methods may reveal longer-term trends or more detailed financial profiles. Finally, experimenting with different thresholds for the recorded behavioral variables could help assess the sensitivity of our findings.

References

[CFPB Codebook](#)

[Financial Well-being Survey Data](#)